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Calculus I (MATH 2200)
Spring 2020
Wednesday, 04/01/2020
Quiz 8: Section 3.11

Section 3.11

5 pts

1. Sand falls from an overhead bin, accumulating in a conical pile with a radius that is always three times its height. Find a formula for $\dfrac{dh}{dt}$.
(Hint: The volume of a cone is $V = (1/3)\pi r^2 h$.)

(A) $\dfrac{dh}{dt} = 3\pi h^3 \dfrac{dV}{dt}$ (B) $\dfrac{dh}{dt} = \dfrac{9h}{\pi}\dfrac{dV}{dt}$ (C) $\dfrac{dh}{dt} = \dfrac{V}{9\pi h^2}$ **(D)** $\dfrac{dh}{dt} = \dfrac{dV}{dt}\dfrac{1}{9\pi h^2}$

The three relevant variables are: the volume V, the radius r, and the height h of the sandpile. The aim is to find the rate of change of the height dh/dt at the instant that $h = 11\ \ ft$, given that $dV/dt = 120\ \ ft^3/min$. The basic relationship among the variables is the formula for the volume of a cone, $V = (1/3)\pi r^2 h$. We now use the given fact that the radius is always three times the height. Substituting $r = 3h$ into the volume relationship gives V in terms of h:

$$V = \frac{1}{3}\pi(3h)^2 h = 3\pi h^3.$$

Rates of change are introduced by differentiating both sides of $V = 3\pi h^3$ with respect to t. Using the Chain Rule, we have:

$$\frac{dV}{dt} = 9\pi h^2 \frac{dh}{dt}.$$

Solving for dh/dt, we have

$$\frac{dh}{dt} = \frac{\frac{dV}{dt}}{9\pi h^2}.$$

5 pts 2. Sand falls from an overhead bin, accumulating in a conical pile with a radius that is always three times its height. If the sand falls from the bin at a rate of 120 ft^3/min, approximately how fast is the height of the sand pile changing when the pile is 11 ft high?
(Hint: The volume of a cone is $V = (1/3)\pi r^2 h$.)

(A) 0.042 ft/min (B) 0.035 **ft/min** (C) 420.089 ft/min (D) 7.173 ft/min

Using our answer from question 1, we find dh/dt at the instant that $h = 11 \ \ ft$, given that $dV/dt = 120 \ \ ft^3/min$. Substituting our values, we have

$$\frac{dh}{dt} = \frac{120 \ \ ft^3/min}{9\pi(11 \ \ ft)^2} \approx 0.035 \ \ ft/min.$$

So, at the instant that the sandpile is 10 ft high, the height is changing at an approximate rate of 0.035 ft/min. Notice how the units work out consistently.